Experiment 6 Day 3 - Conversion of Borneol to Camphor

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Chem 2081-007

Introduction

In this experiment, borneol was converted to camphor, the final product of the three-day experiment.

Materials and Methods

First, 0.296 g of the borneol product from day 2 of the experiment was added to a long neck round bottom flask with 0.3 mL of glacial acetic acid. This mixture was then dissolved in 1 mL of acetone. A distillation apparatus was assembled and the flask, with a boiling chip inside, was placed in a 40° C water bath. Once the borneol was fully dissolved in the acetone, 6 mL of bleach was added dropwise at a rate of ½ mL per minute. The reaction was allowed to continue in the water bath for 15 minutes after the final addition of bleach. Once the reaction was complete, the flask was cooled to room temperature, and the solution was transferred to a centrifuge tube. The organic product was extracted twice with two separate 2 mL portions of DCM and placed in a 10 mL beaker. The aqueous waste was transferred to another beaker, and the organic material was returned to the centrifuge tube and neutralized with 2 mL of saturated sodium bicarbonate. The product was again extracted from the centrifuge tube, and the waste remaining was added to the waste beaker. The product was returned to the centrifuge tube and was the 2 mL of 5% sodium bisulfite and then 2 mL of deionized water. The organic product and waste were separated a final time, and the organic product was dried in the centrifuge tube with anhydrous sodium sulfate. The organic solution was decanted onto a pre-weighed watch glass, and the DCM was evaporated over a water bath. The watch glass was weighed again the camphor crystals on top, and the percent yield was calculated. All the waste and remaining compounds were properly disposed of, and all the glassware used was properly cleaned and stored in the drawer.

Results

Mass of clean watch glass (g)	30.476
Mass of watch glass + camphor (g)	30.675
Mass of borneol (g)	0.199

Calculations

$$30.675\ g\ watch\ glass\ with\ camphor - 30.476\ g\ clean\ watch\ glass$$

$$= 0.199\ g\ camphor\ crystals$$

$$\frac{(0.296\ g\ borneol)}{154.25\frac{g}{mol}} = 0.0019\ mol\ borneol$$

$$0.0019\ mol\ borneol \times 152.23\frac{g}{mol}\ camphor = 0.292\ g\ theoretical\ yield\ camphor$$

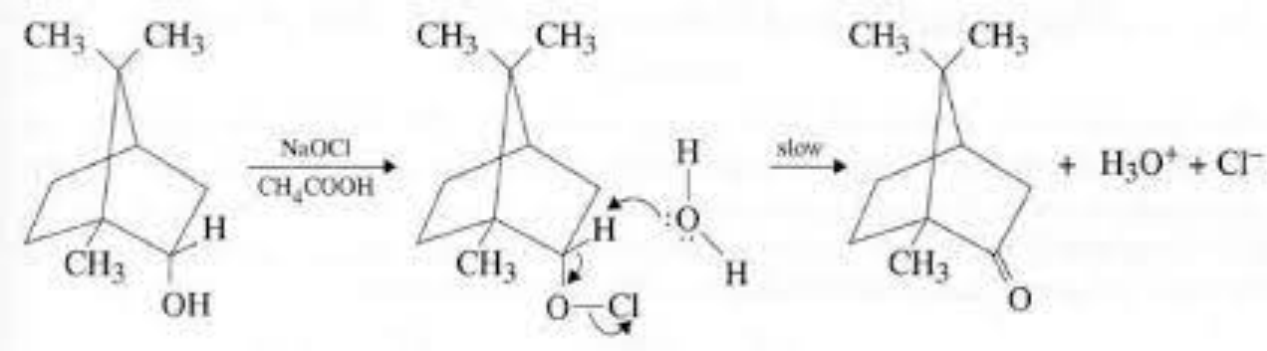
$$\frac{0.199\ g\ actual\ yield\ camphor}{0.292\ g\ theoretical\ yield\ camphor} \times 100\% = 68.2\%\ yield\ camphor$$

Discussion

In this experiment, the borneol product, an alcohol, obtained in day 2 of the experiment was oxidized to camphor, a ketone, with the use of bleach. The mechanism below shows how the reaction proceeds to form camphor in the presence of both bleach and water. In the reaction mixture, the acetone, bleach, and acetic acid are all dissolved in water, meaning the reaction mixture had a density of 1 g/mL. Once the reaction was complete, the organic product, camphor, was dissolved in DCM, which has a density of 1.33 g/mL. Because the organic material was dissolved a solution denser than water, the organic layer was the bottom layer in this experiment.

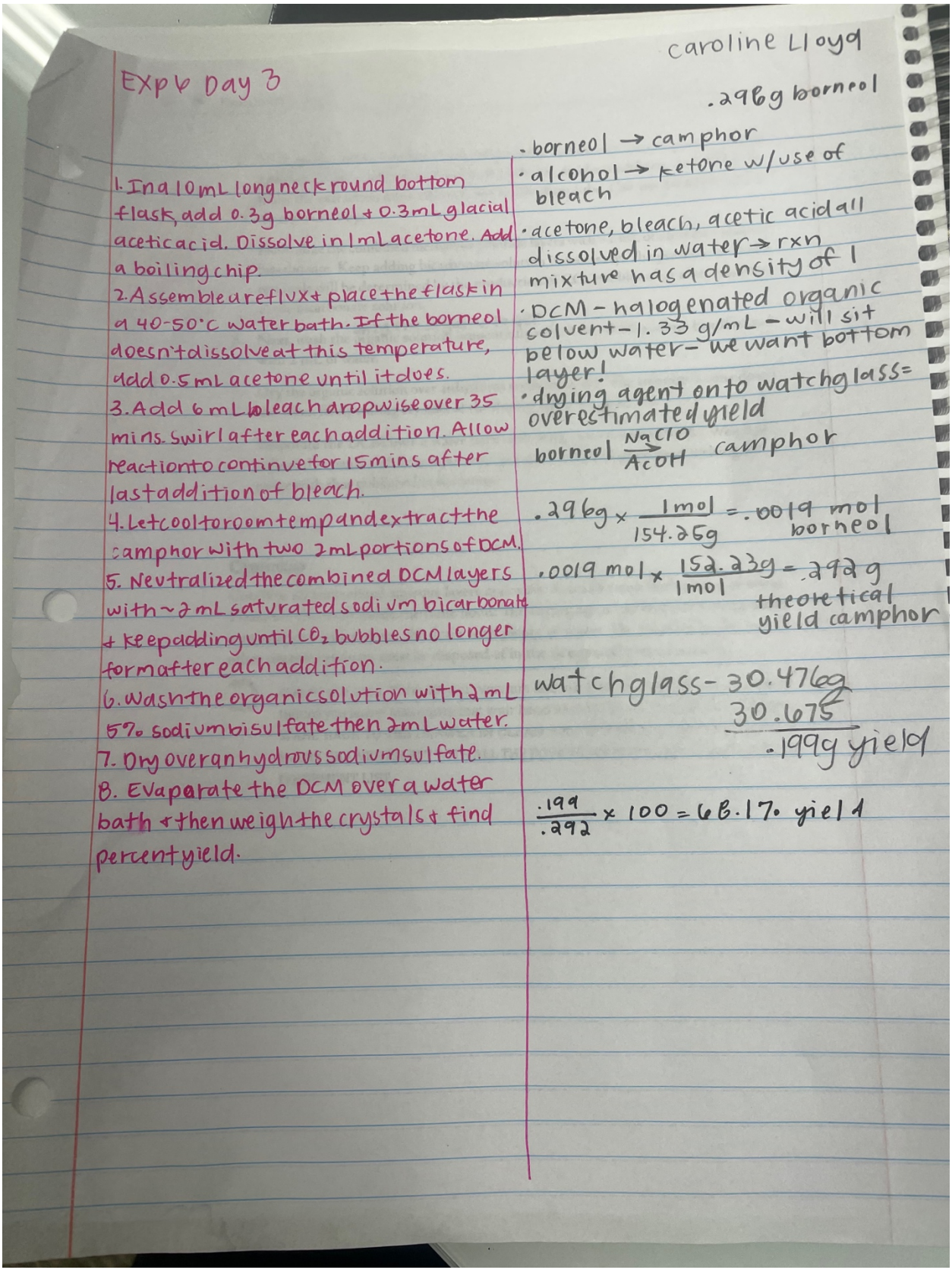
There are several possible sources of error in this experiment. The first, and most likely, source of error is having an impure starting material. Using an impure collection of borneol to

begin the experiment would lead to an underestimated percent yield of camphor because the amount of true borneol was inaccurately calculated. Additionally, in any extraction of organic product from waste, a separatory funnel should always be used to ensure complete separation of the two layers. Without the use of a separatory funnel, it is impossible to confirm that the organic and aqueous layers are completely separated and extracted from each other, and this can potentially lead to an overestimated or an underestimated yield of camphor.



References

Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry 2 Lab Manual [lab manual]. Auburn (Al): Auburn University.



Exp 6 Day 3

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.296g borneol

1. In a 10mL long neck round bottom flask, add 0.3g borneol + 0.3mL glacial acetic acid. Dissolve in 1mL acetone. Add a boiling chip.
2. Assemble reflux + place the flask in a 40-50°C water bath. If the borneol doesn't dissolve at this temperature, add 0.5mL acetone until it does.
3. Add 6mL bleach dropwise over 35 mins. Swirl after each addition. Allow reaction to continue for 15 mins after last addition of bleach.
4. Let cool to room temp and extract the camphor with two 2mL portions of DCM.
5. Neutralized the combined DCM layers with ~2mL saturated sodium bicarbonate + keep adding until CO_2 bubbles no longer form after each addition.
6. Wash the organic solution with 2mL 5% sodium bisulfate then 2mL water.
7. Dry over anhydrous sodium sulfate.
8. Evaporate the DCM over a water bath + then weigh the crystals + find percent yield.

- borneol → camphor
- alcohol → ketone w/use of bleach
- acetone, bleach, acetic acid all dissolved in water → rxn mixture has a density of 1
- DCM - halogenated organic solvent - 1.33 g/mL - will sit below water - we want bottom layer!
- drying agent onto watchglass = overestimated yield

borneol $\xrightarrow[AcOH]{NaClO}$ camphor

$.296g \times \frac{1mol}{154.25g} = .0019\ mol$ borneol

$.0019\ mol \times \frac{152.23g}{1mol} = .292g$ theoretical yield camphor

watchglass - 30.476g

30.675

.199g yield

$\frac{.199}{.292} \times 100 = 68.1\%$ yield

Lab Performance

Exp 6 D3

Caroline Lloyd | 007 | 402 | T

Name: | *Section:* | *Room:* | *Hood:*

Wearing personal protection equipment—this evaluation records your consistency with wearing the appropriate clothes for lab (long pants, closed toe shoes, and tops with sleeves) as well as the required laboratory safety equipment of apron, goggles, and gloves. ○ ○ ○ ○ ~~○~~ (5pts)

Safe working habits—as the GTAs do the lab rotations, they will record your work ethic and whether you follow our safety rules and regulations concerning tied hair, tied apron, safety goggles on at all times, and no food and drinks. ○ ○ ○ ○ ~~○~~ (5pts)

Following our laboratory safety technique work procedures—evaluation of operation of our laboratory equipment—water sinks in hood (1pt), vacuum and air lines (1pt), sash position (1pt), and heating elements (1pt). It is important that after every experiment you shut down the hood and make it ready for the section after yours (5pts). ○(1pt) ○(1pt) ○(1pt) ○ (1pt) ~~○~~ (5pts)

Glassware operation—safe handling of our glassware equipment. This means cleaning (1pt), maintaining the drawer inventory (5pts), and proper experimental setup (5pts). ○(1pt) ○ (5pts) ~~○~~ (5pts)

Working independently or equal partner participation during the entire lab. ○ ○ ○ ○ ~~○~~ (5pts)

Good work behavior—you work in a dynamic environment; therefore, creating distractions through noise and disobedience will not be tolerated. No cell phone use, listening to loud music, or working/distracting at a non-assigned workspace. ○ ○ ○ ○ ~~○~~ (5pts)

Maintain clean and organized personal space. ○ ○ ○ ○ ~~○~~ (5pts)

GTA notes:

GTA signature: _____ *Final Performance Grade:* [ ]